

Lec 5-2 . Oct 12 (Sun), 2025 .

- concavity vs secant lines
 - Concavity vs tangent lines
 - Applied optimization (4.5)
 - Newton's method (4.6)
- (4.4 extended, not in the book)
need to know the facts (theorems);
proofs are optional .

• Summary for Local Extrema

First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f , and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across c from left to right,

1. if f' changes from negative to positive at c , then f has a local minimum at c ;
2. if f' changes from positive to negative at c , then f has a local maximum at c ;
3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c .

4.4.5

THEOREM 4.4.5 — Second Derivative Test for Local Extrema

on an open interval that contains $x = c$.

Suppose f'' ~~is continuous~~ ^{exists}

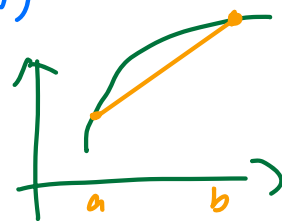
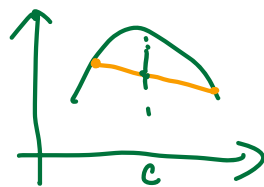
1. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

"Concavity" vs "Secant & Tangent Lines"

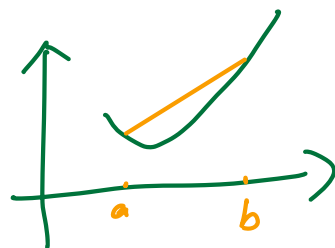
Thm [Concavity and Secant Lines]

. Let f be cont' on $[a, b]$ and differentiable on (a, b) .

(i) If f is concave down on (a, b) , then the graph of f lies above the secant line joining $(a, f(a))$ and $(b, f(b))$ on (a, b) .



(ii) If f is concave up on (a, b) , then the graph of f lies below the secant line joining $(a, f(a))$ and $(b, f(b))$ on (a, b) .



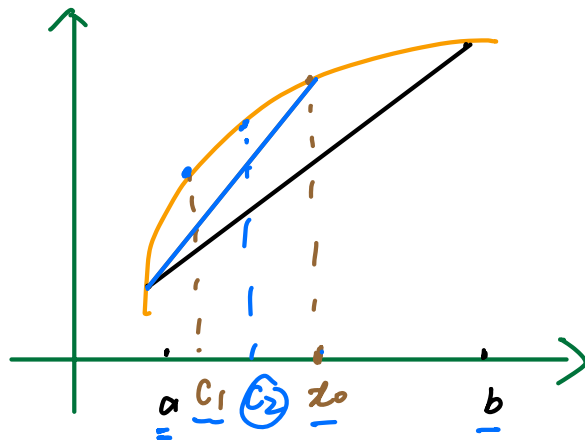
Pf [Optional]

(i) Step 1: analyze the problem

We are going to prove

$$f(x) > g(x) \quad \forall x \in (a, b).$$

where $g(x) \triangleq f(a) + \frac{f(b)-f(a)}{b-a}(x-a), \quad x \in (a, b).$



$\therefore$ It is equivalent to show

$$f(x_0) - f(a) > \frac{f(b)-f(a)}{b-a}(x_0-a), \quad \forall x_0 \in (a, b).$$

Step 2. Fix $x_0 \in (a, b)$ and use MVT.

Fix $x_0 \in (a, b)$. By MVT, $\exists c_1 \in (a, x_0)$.

$$\text{s.t. } \frac{f(x_0) - f(a)}{x_0 - a} = f'(c_1).$$

We are going to show

$$f'(c_1) > \frac{f(b)-f(a)}{b-a},$$

$$\text{i.e., } f(b) < f(a) + (b-a)f'(c_1) \quad (*)$$

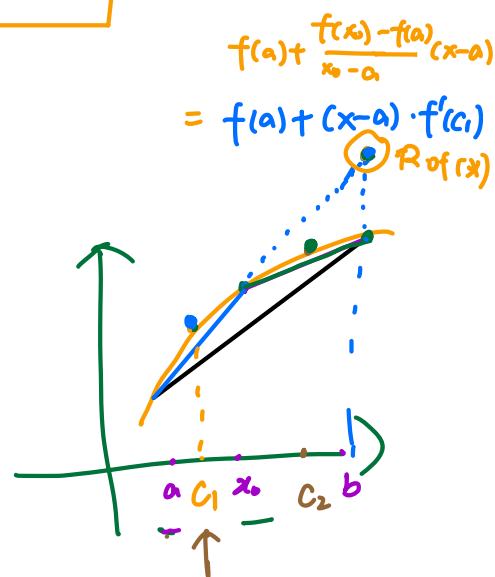
$$y(x) = f(a) + f'(c_1)(x-a)$$

$$R \text{ of } (*) = y(x) \Big|_{x=b}$$

Step 3. Divide into two intervals.

$$\underline{f(b)} = f(a) + \frac{f(x_0)-f(a)}{x_0-a}(x_0-a) + \frac{f(b)-f(x_0)}{b-x_0}(b-x_0)$$

$$= f(a) + f'(c_1)(x_0-a) + \frac{f(b)-f(x_0)}{b-x_0}(b-x_0)$$



For $[x_0, b]$, using MVT again, there exists $c_2 \in (x_0, b)$,

$$\text{s.t.} \quad \frac{f(b) - f(x_0)}{b - x_0} = f'(c_2).$$

$\therefore f'$ is decreasing on (a, b) and $c_1 < c_2$

$$\therefore f'(c_1) > f'(c_2).$$

Step 4. Two derivatives.

$$\begin{aligned} \therefore f(b) &= f(a) + f'(c_1)(x_0 - a) + \frac{f(b) - f(x_0)}{b - x_0}(b - x_0) \\ &= f(a) + f'(c_1)(x_0 - a) + f'(c_2)(b - x_0) \\ &< f(a) + f'(c_1)(x_0 - a) + f'(c_1)(b - x_0) \\ &= f(a) + f'(c_1)(b - a) = \text{RHS of } (*) \end{aligned}$$

$\therefore$ we prove $(*)$.

(ii) Similar.

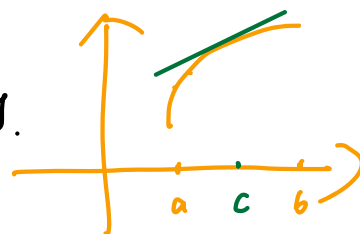
□.

Thm (Concavity and Tangent lines).

Let f be continuous on $[a, b]$ and differentiable on (a, b) .

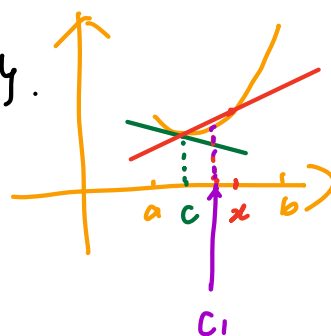
(i) If f is concave down on (a, b) , then for any $c \in (a, b)$, the tangent line to $y = f(x)$ at c lies above the graph of $y = f(x)$, i.e.,

$$f(x) < f(c) + f'(c)(x - c), \quad \forall x \in (a, b) \setminus \{c\}.$$



(ii) If f is concave up on (a, b) , then for any $c \in (a, b)$, the tangent line to $y = f(x)$ at c lies below the graph of $y = f(x)$, i.e.,

$$f(x) > f(c) + f'(c)(x - c), \quad \forall x \in (a, b) \setminus \{c\}.$$



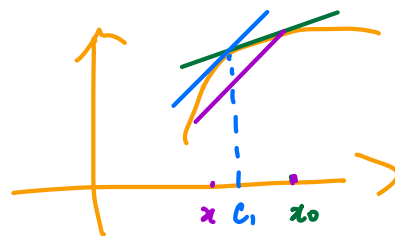
[optional]

Pf: Step 1: Analyze the problem.

① We are going to prove: For fixed $x_0 \in (a, b)$,

$$f(x) < f(x_0) + f'(x_0)(x - x_0) \quad \forall x \in (a, b) \setminus \{x_0\}$$

$$\Leftrightarrow f(x) - f(x_0) < f'(x_0)(x - x_0) \quad \forall x \in (a, b) \setminus \{x_0\}.$$



Step 2. MVT!

For $x \in (a, x_0)$, we are going to prove

$$\frac{f(x) - f(x_0)}{x - x_0} > f'(x_0) \quad (*)$$

We use MVT: $\exists c_1 \in (x, x_0)$, s.t.

$$f'(c_1) = \frac{f(x) - f(x_0)}{x - x_0}.$$

$\therefore f'$ is decreasing and $c_1 < x_0$

$$\therefore f'(c_1) > f'(x_0)$$

$$\therefore f'(c_1) = \frac{f(x) - f(x_0)}{x - x_0} > f'(x_0), \text{ which implies } (*).$$

For $x \in (x_0, b)$, we are going to prove

$$\frac{f(x) - f(x_0)}{x - x_0} < f'(x_0). \quad (**).$$

We use MVT: $\exists c_2 \in (x_0, x)$, s.t.

$$f'(c_2) = \frac{f(x) - f(x_0)}{x - x_0}.$$

$\therefore f'$ is decreasing and $c_2 > x_0 \therefore f'(c_2) < f'(x_0)$

$$\therefore f'(c_2) = \frac{f(x) - f(x_0)}{x - x_0} < f'(x_0), \text{ which implies } (**). \quad \square$$

(ii) Similar.

$\square$.

• Applied optimization

Eg2. A cylindrical can with ^{two} lids is to be made to hold 1 L of oil. $\overset{=1000 \text{ mL}}{\text{1 L}}$

Find the dimensions that will minimize the cost of the metal to manufacture the can.

Sol: Goal: would like to minimize

$$f = 2 \cdot \pi r^2 + 2\pi r \cdot h$$

subject to $\pi r^2 \cdot h = 1000$, where $r > 0$.

② Write down the function:

$$\text{Let } f(r) = 2\pi r^2 + 2\pi r \cdot \frac{1000}{\pi r^2}$$

$$= 2\pi r^2 + \frac{2000}{r}, \quad r > 0$$

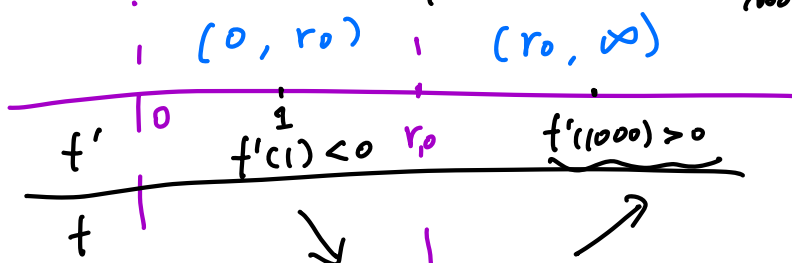
③ Solve the local min.

$$f'(r) = 4\pi r - \frac{2000}{r^2}, \quad r > 0$$

$$\text{Let } f'(r) = 0 \Rightarrow r_0 = \left(\frac{500}{\pi}\right)^{\frac{1}{3}}$$

First order test

Analyse:



$$f'(r_0) = 4\pi - \frac{2000}{r_0^2} < 0$$

$$f'(1000) = 4\pi \times 1000 - \frac{2000}{1000^2} > 0$$

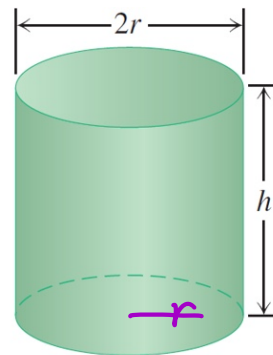


FIGURE 4.36 This one-liter can uses the least material when $h = 2r$ (Example 2).

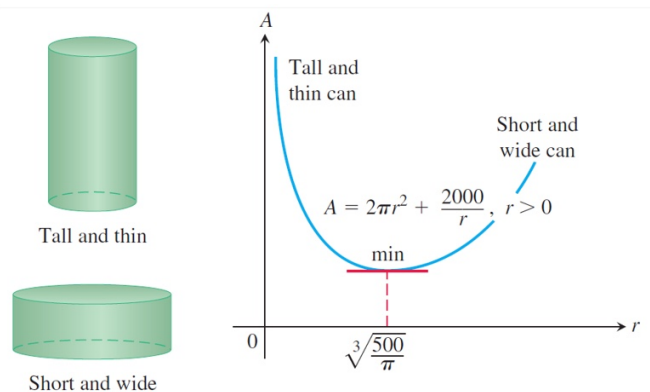


FIGURE 4.37 The graph of $A = 2\pi r^2 + 2000/r$ is concave up.

OR Second order Test

$$f''(r) = 4\pi + \frac{4000}{r^3} > 0$$

$$\Rightarrow f''(r_0) > 0.$$

$\Rightarrow f$ has a local min at $c = \left(\frac{500}{\pi}\right)^{1/3}$

④ Solve the global min.

As f is $\searrow$ on $(0, r_0)$ and $\nearrow$ on (r_0, ∞) ,

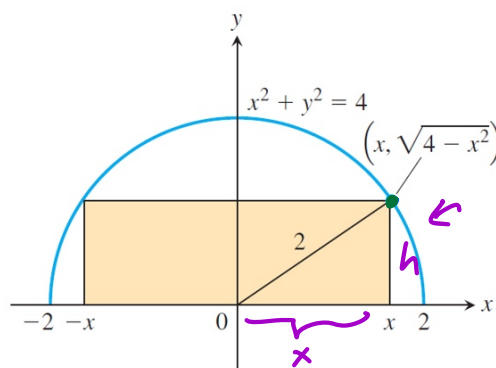
we know f attains its absolute minimum at r_0 .

$$\therefore h_0 = \frac{1000}{\pi r_0^2} = \frac{1000}{\pi \left(\frac{500}{\pi}\right)^{2/3}} = 2 \left(\frac{500}{\pi}\right)^{1/3} (= 2 \cdot r_0) \quad \square.$$

$$V = \pi r_0^2 \cdot h_0 = \dots$$

Ex 2. A rectangle is to be described in a semicircle of radius 2.

What is the largest area the rectangle can have?



$h, x > 0$

FIGURE 4.38 The rectangle inscribed in the semicircle in Example 3.

Sol: Step 1: write down the objective!

Would like to maximize $2 \cdot x \cdot h$

$$\text{s.t. } x^2 + h^2 = 4, \text{ where } x > 0.$$

Step 2: write down the function!

$$2 \cdot \sqrt{2} \cdot \sqrt{4-2} = 4$$

Let $f(x) = 2x \cdot \sqrt{4-x^2}$, $x \in [0, 2]$.

$$f(0) = 0 \quad f(2) = 0.$$

Step 3: solve the local max!

$$f'(x) = 2\sqrt{4-x^2} + 2x \left(\frac{1}{2}\right) (4-x^2)^{-\frac{1}{2}} \cdot (-2x)$$

$$= \dots$$

$$= \dots$$

$$= 4 \frac{2-x^2}{\sqrt{4-x^2}}, \quad x \in (0, 2)$$

$$4 \cdot \frac{2-1}{\sqrt{1}} > 0$$

Let $f'(c) = 0 \Rightarrow$ we have $c = \sqrt{2}$.

$$\frac{2-1.5^2}{\sqrt{1}} < 0$$

Analyse:

first derivative test

or SDT

	$(0, \sqrt{2})$	$(\sqrt{2}, 2)$
f'	$f'(1) > 0$ +	$f'(1.5) < 0$ -
f	$\nearrow$	$\searrow$

$$f''(c) < 0$$

Hence, f has a local max at $\sqrt{2}$.

Step 4. Solve the absolute max!

As f is $\nearrow$ on $(0, \sqrt{2})$, $\searrow$ on $(\sqrt{2}, 2)$

$\therefore f$ has a global max at $\sqrt{2}$.

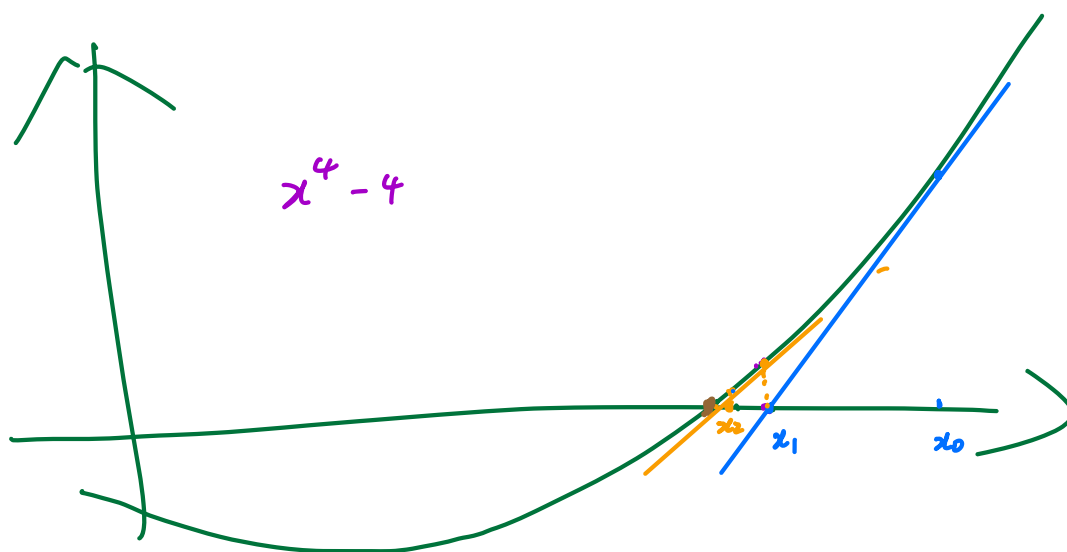
□

Note that $f(0) = 0$, $f(2) = 0$. $f(\sqrt{2}) = 4$.

indeed f has an absolute max at $\sqrt{2}$.

• Newton's method

Goal: to find the root of a known function f .



General procedure:

1. Start with a point x_0 "near" the root r .

2. Let L_0 be the tangent line at $x = x_0$.

If $f'(x_0) \neq 0$, L_0 will intersect the x-axis at some point; call it x_1 .

3. For $i = 1, 2, 3, \dots$.

Let L_i be the tangent line at $x = x_i$.

If $f'(x_i) \neq 0$, L_i will intersect the x -axis at some point; call it x_{i+1} .

4. Hopefully, as i increases, x_i will be closer and closer to the root r .

Specifically,

① Starting with x_0 , we have

$$L_0(x) = f(x_0) + f'(x_0)(x - x_0).$$

Letting $L_0(x_1) = 0$, we have

$$f(x_0) + f'(x_0)(x_1 - x_0) = 0 \Rightarrow x_1 - x_0 = \frac{-f(x_0)}{f'(x_0)} \Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

② Starting with x_i , we have

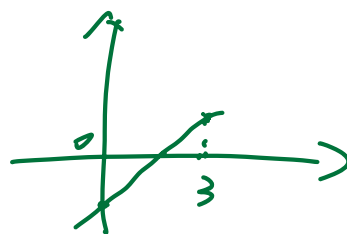
$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}, \quad i = 1, 2, 3, \dots$$

e.g. Try to compute the root of $f(x) = x^3 - 2x - 5$.

Sol: Step 1. Find a starting point x_0 !

Note that $f(0) = -5$, $f(3) = 3^3 - 6 - 5 > 0$.

As f is continuous on $[0, 3]$,

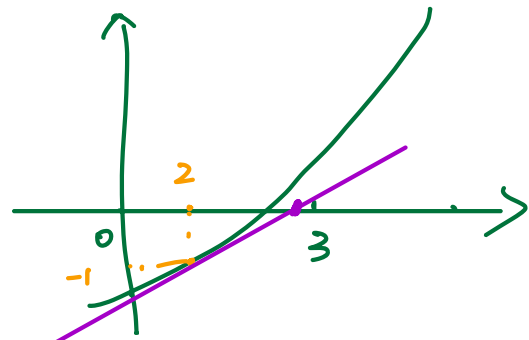


by the intermediate value thm, there exists some $r \in (0, 3)$ s.t. $f(r) = 0$.

$\therefore$ we choose $x_0 = 2$ ($\in (0, 3)$)

$$\text{and compute } f(x_0) = 2^3 - 2 \times 2 - 5 = -1.$$

Indeed quite close to the root!



Step 2. Compute x_1 .

$$\text{Compute: } f'(x) = 3x^2 - 2 \Rightarrow f'(x_0) = 3 \cdot 2^2 - 2 = 10.$$

Write down the tangent line L_0 :

$$\begin{aligned} L_0(x) &= f(x_0) + f'(x_0)(x - x_0) \\ &= -1 + 10(x - 2) \end{aligned}$$

$\therefore$ Letting $L_0(x_1) = 0$, we have

$$10(x_1 - 2) - 1 = 0 \Rightarrow x_1 = 2 + \frac{1}{10} = 2.1.$$

Step 3. Compute x_2 .

Write down the tangent line L_1 :

$$\begin{aligned} L_1(x) &= f(x_1) + f'(x_1)(x - x_1) \\ &= f(2.1) + f'(2.1)(x - 2.1). \end{aligned}$$

$\therefore$ Letting $L_1(x_2) = 0$, we have

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.1 - \frac{f(2.1)}{f'(2.1)} = 2.094568 \dots$$

Step 4. Compute $x_3, x_4, x_5, \dots$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.094551 \dots$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 2.094551 \dots$$

$\therefore$ The point slightly change

$\therefore 2.094551$ can be taken as an approximation of r .

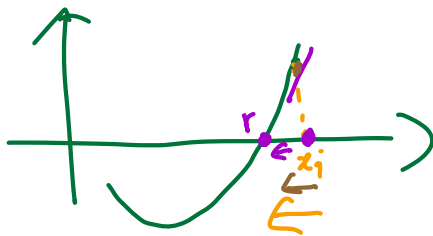


Intuition

$$\textcircled{1} \quad x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}, \quad i = 0, 1, 2, 3, \dots$$

"compensation effect" : between the value and the trend !

$f'(x_i) > 0$



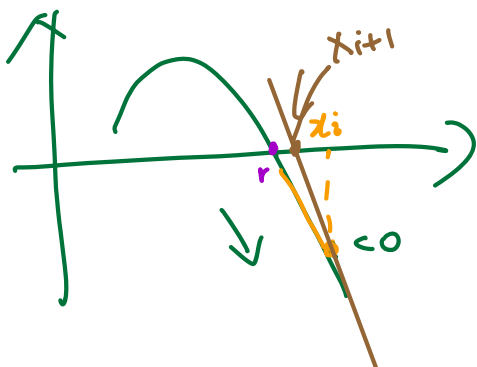
here $f(x_i) > 0 = f(r)$,

implying that f is "increasing" on (r, x_i) and $f'(x_i) > 0$.

Then $\frac{f(x_i)}{f'(x_i)} > 0$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

You need to deduct something to approach $f(r) = 0$!

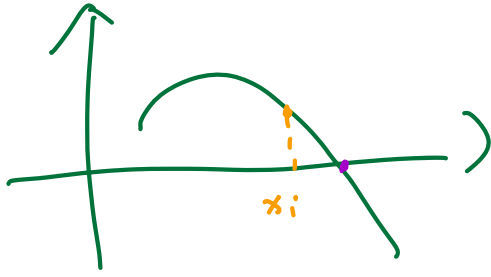


here $f(x_i) < 0 = f(r)$

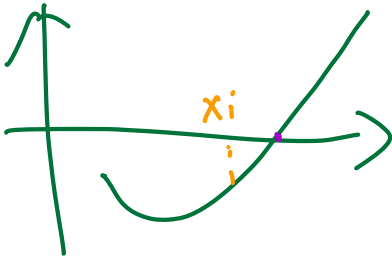
implying f is "decreasing" on (r, x_i) and $f'(x_i) < 0$

Then $\frac{f(x_i)}{f'(x_i)} > 0$. Also deduct !

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$



$$\frac{f(x_i) > 0}{f'(x_i) < 0}$$



$$\frac{f(x_i) < 0}{f'(x_i) > 0}$$

② When close to r , we have $f(x_i) \approx f(r) = 0$.

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \approx 0$$

$\therefore x_i, i=1, 2, \dots, n$, converges to r !

[$x_i, i=1, 2, \dots, n$, does not change a lot and "stop" at some point r]

$$\text{[For } r, f(r) = 0 \Rightarrow r - \frac{f(r)}{f'(r)} = r !$$

- Newton's method via codes (e.g., MATLAB)

$$f = @(x) x^4 - 4;$$

$$df = @(x) 4x^3;$$

$$x = 1; \text{ (this is your initial guess } x_0)$$

$$x = x - f(x)/df(x);$$

$$x = x - f(x)/df(x);$$

....

- Newton's method does not perform well

if $f'(x_i) \approx 0$.

• Antiderivatives

Def: If $F'(x) = f(x)$ for all x in an interval I , then

F is said to be an **antiderivative** of f on I .

By a corollary of the MVT, we have:

Thm 4.7.8. If F is an antiderivative of f on I , then all the antiderivatives of f on I are $F + C$, where C is an arbitrary constant.

$$\cdot G' = f$$

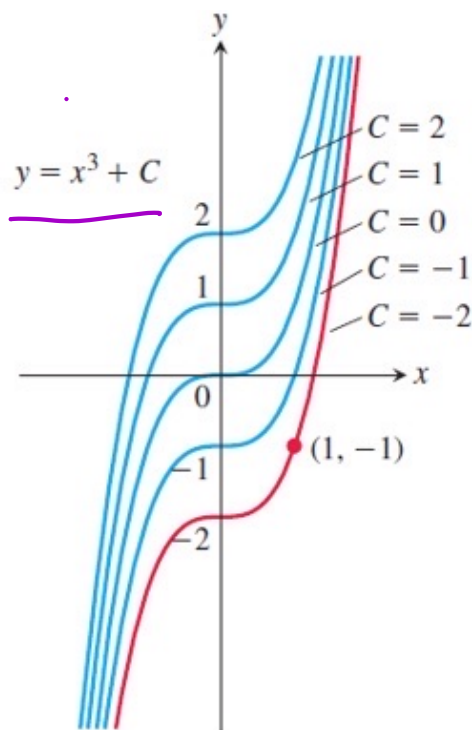
$$\cdot F' = f$$

$$\Rightarrow (G - F)' = 0$$

$$\stackrel{\text{MVT}}{\Rightarrow} G - F \equiv \text{const}$$

$$\Rightarrow G = F + C,$$

for some $C \in \mathbb{R}$



$$F(x) = x^3 + C, \text{ where } C \text{ is any real number.}$$

$$F(x) = x^3$$
$$f(x) = 3x^2$$

All antiderivatives are "parallel".

Function	General antiderivative
1. x^n	$\frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$
2. $\sin kx$	$-\frac{1}{k}\cos kx + C$
3. $\cos kx$	$\frac{1}{k}\sin kx + C$
$\frac{1}{\cos^2(kx)} \neq \sec^2 kx$	$\frac{1}{k}\tan kx + C$

Some common formulae.

e.g. Find all antiderivatives of $f(x) = 3\sqrt{x} + \sin 2x$.

Sol': $F(x) = 2x^{\frac{3}{2}} + (-\frac{1}{2})\cos 2x + C,$

where C is any real number.

□